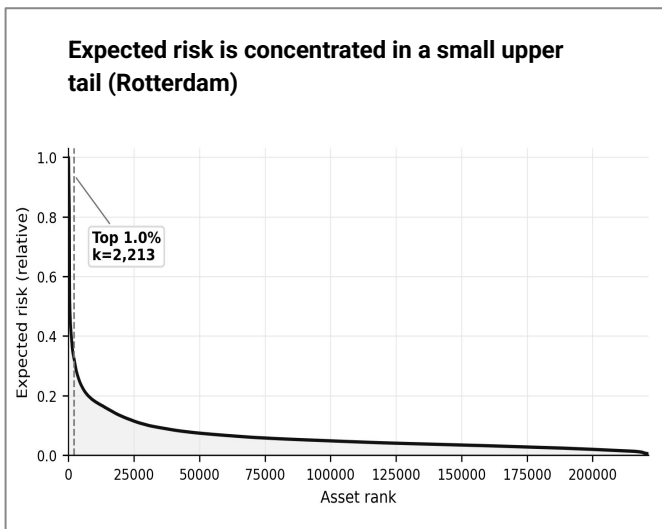


When uncertain inputs become hard decisions

Planning and prioritisation often rely on fixed thresholds and rankings.

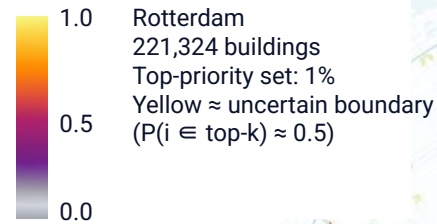
But uncertain inputs can produce unstable decisions near the decision boundary.

Habnetic estimates posterior decision stability to identify the small subset of assets whose prioritisation is genuinely uncertain.



CITYWIDE OVERVIEW - ROTTERDAM

Top-k membership probability
($k = 2,213$)



TAKEAWAY

**Risk is concentrated.
Decision uncertainty is even more concentrated.**

Prioritisation itself can be treated as a probabilistic object

Habnetic propagates uncertainty in exposure and hazard indicators through a Bayesian risk model and into the decision rule.

The output is not a risk score or a fixed ranking, but posterior decision metrics that quantify the stability of prioritisation decisions.

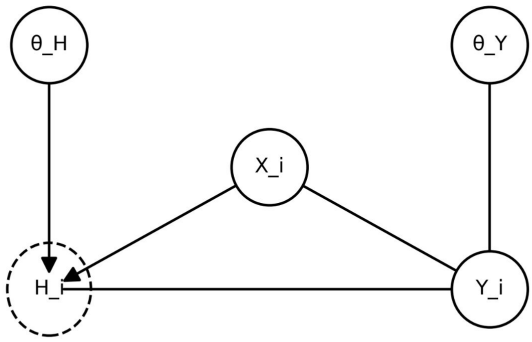
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PROBABILISTIC FRAMEWORK

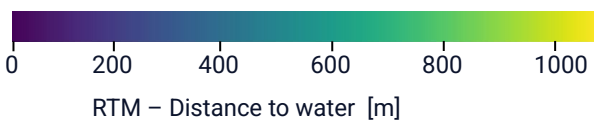
Exposure and hazard indicators are combined within a Bayesian model.

Output:
Posterior decision metrics

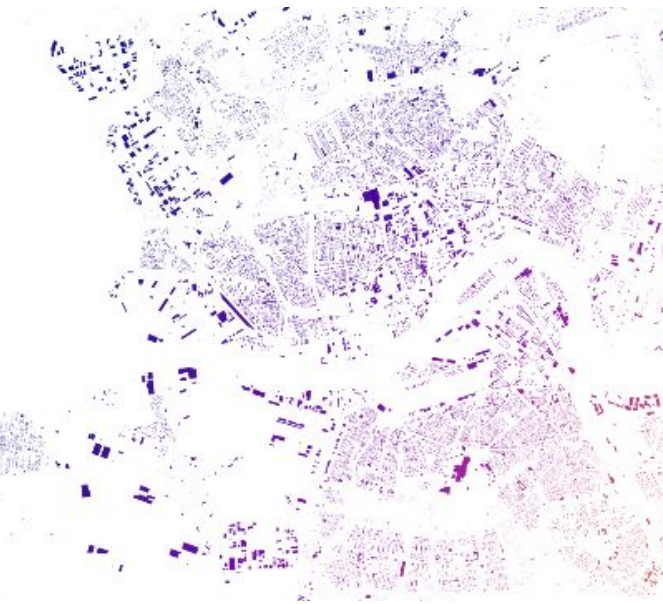
X_i Exposure indicators
 I_i Hazard proxy
 Y_i Impact measure
 θ_H Hazard parameters
 θ_Y Impact parameters



EXPOSURE INDICATOR



HAZARD INDICATOR



TAKEAWAY
Inputs become posterior decision metrics.

Habnetic derives posterior decision metrics rather than deterministic rankings.



Where does decision uncertainty concentrate?

Posterior decision stability reveals where prioritisation is robust and where decisions remain uncertain. Most assets remain stable, while uncertainty concentrates near the prioritisation boundary.

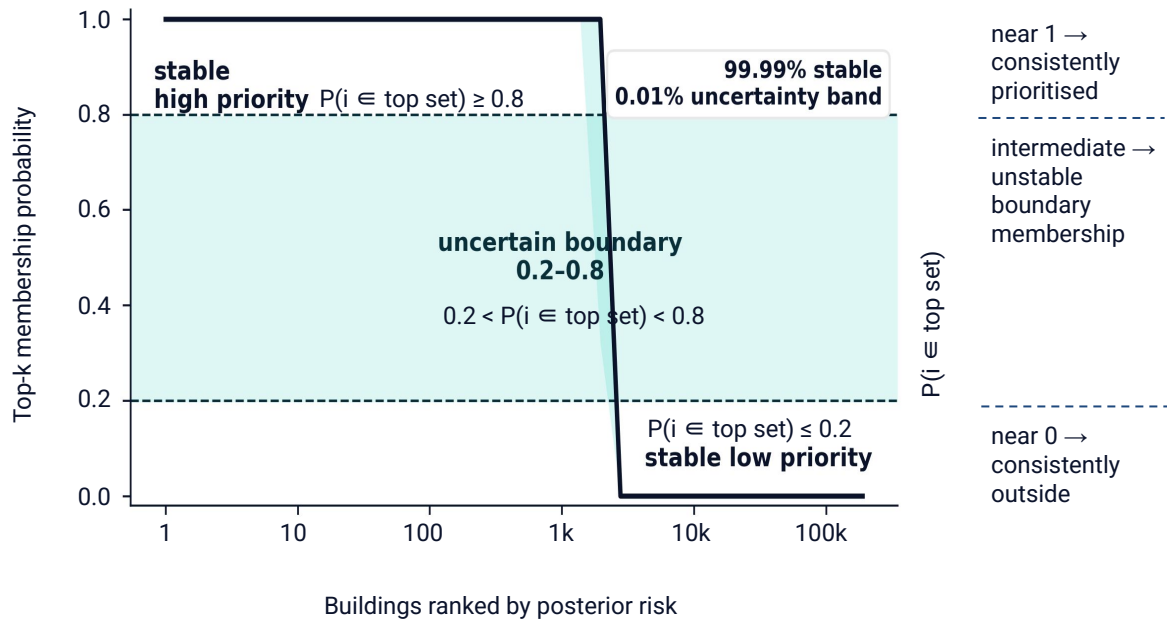


Habnetic reveals that only a small subset of boundary assets remains uncertain.

Top-k membership probability measures how consistently each asset belongs to the selected priority set across posterior simulations.

POSTERIOR STABILITY STRUCTURE

Rotterdam baseline • top-priority set = 1.0% of buildings • k = 2,213



TAKEAWAY

99.99% of buildings remain stable. Decision uncertainty concentrates near a narrow decision boundary.



Only boundary assets remain uncertain.



Review only boundary decisions.



Prioritise robust decisions.



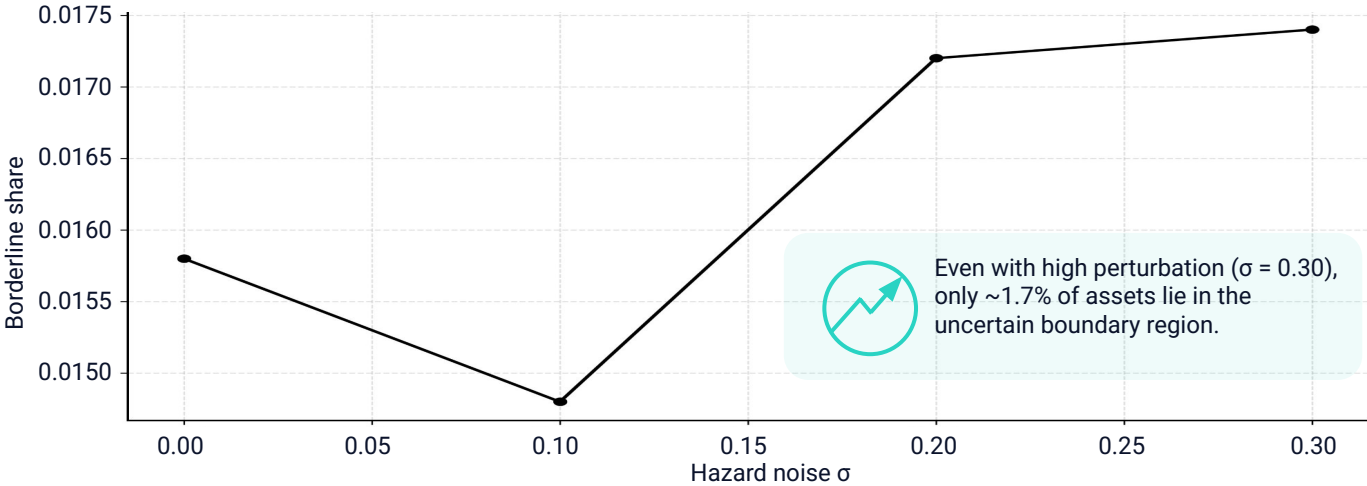
Decision stability remains robust under perturbation

Hazard perturbation increases uncertainty, but it remains concentrated near a narrow decision boundary rather than spreading across the system.

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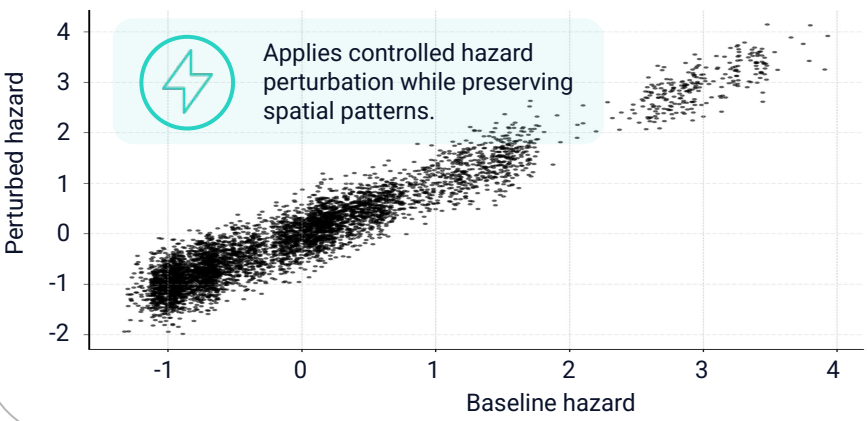
BOUNDARY SHARE UNDER HAZARD PERTURBATION

Rotterdam · top-priority set = 1.0% of buildings · $k = 2,213$



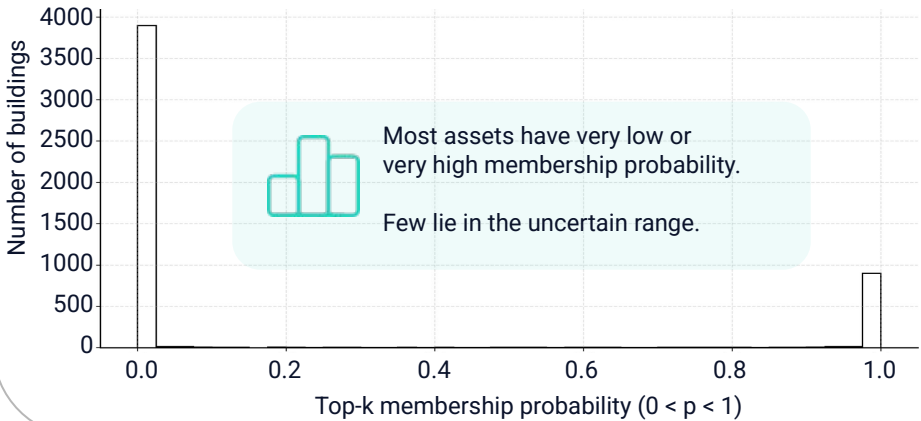
EXAMPLE: HAZARD PERTURBATION ($\sigma = 0.10$)

Building-level perturbation example



POSTERIOR DECISION PROBABILITIES REMAIN CONCENTRATED

Distribution of top-k membership probabilities ($k = 1000$)



TAKEAWAY

Decision uncertainty remains localised under substantial hazard perturbation.



Hazard perturbation increases variation...



...but instability stays concentrated...



...preserving confidence in top-set decisions.

Decision-stability structure persists across cities

The same probabilistic specification preserves the overall decision stability structure across cities, while distributional shift produces only moderate broadening of the uncertainty boundary.

The uncertainty boundary remains narrow in every city.

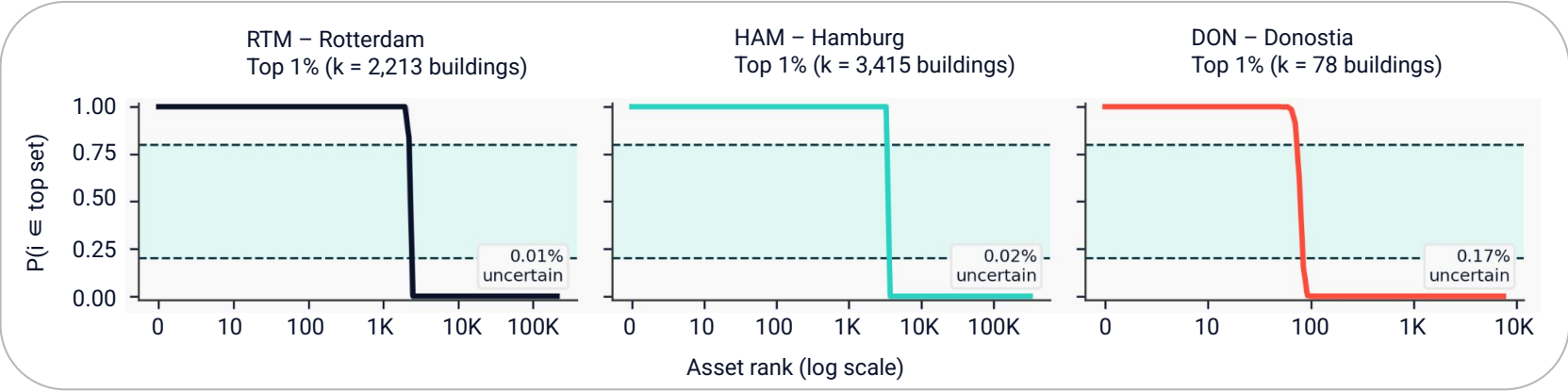


TAKEAWAY

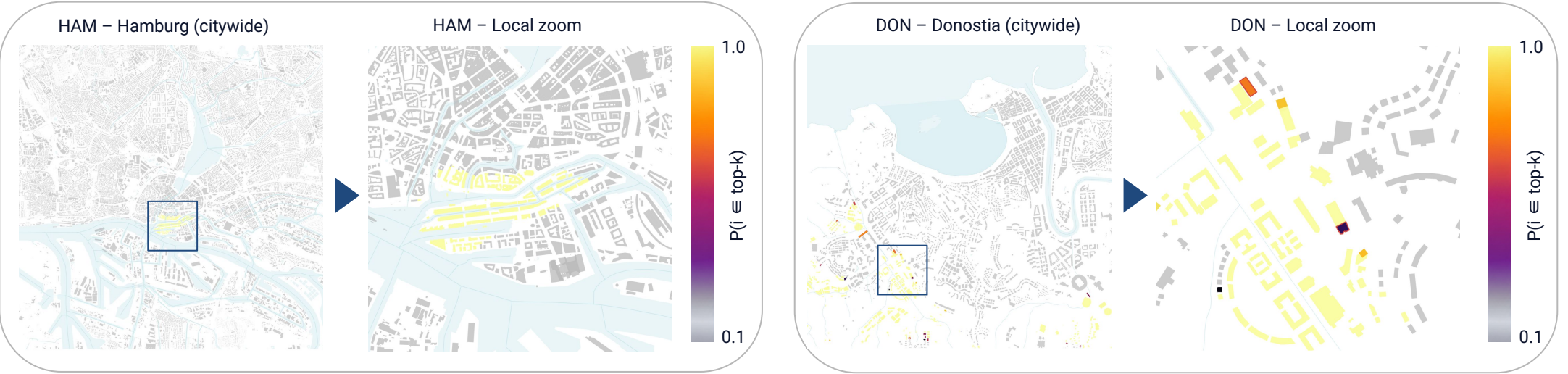
Decision-stability structure remains consistent across cities.
Uncertainty remains concentrated near a narrow boundary.

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STABILITY STRUCTURE BY CITY (TOP-SET = 1.0% OF BUILDINGS)



UNCERTAINTY REMAINS LOCALISED IN EVERY CITY





TAKEAWAY

Decision-stability structure remains consistent under fixed-specification transfer



Stable classifications are preserved.



Uncertain boundary broadens modestly.



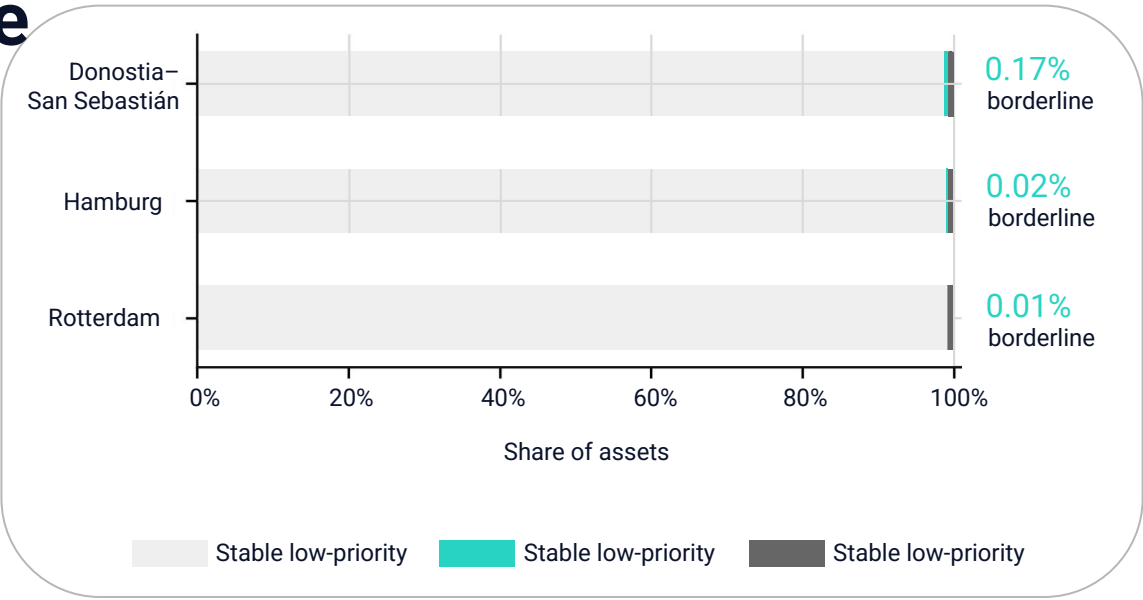
Uncertainty stays localised near the decision boundary.

Do not review uncertainty everywhere. Review where it can change the decision.

Posterior decision stability supports uncertainty-aware prioritisation, targeted review, and robustness assessment under distributional shift.

Rather than reviewing every asset, attention can be focused on the small subset of boundary assets whose prioritisation remains uncertain.

MOST PRIORITISATION DECISIONS REMAIN STABLE ACROSS CITIES



TAKEAWAY

The uncertain boundary is consistently very small, from 0.01% to 0.17% of assets across cities.

FROM REVIEWING EVERYTHING (INEFFICIENT)



TO FOCUSING THE BOUNDARY (EFFECTIVE)



TO ACTING WITH CONFIDENCE (RELIABLE)



TAKEAWAY

Decision uncertainty is concentrated in a small boundary that can be reviewed explicitly.



Focus review effort on the boundary, not the entire portfolio.



Targeted review concentrates effort where classification is genuinely uncertain.



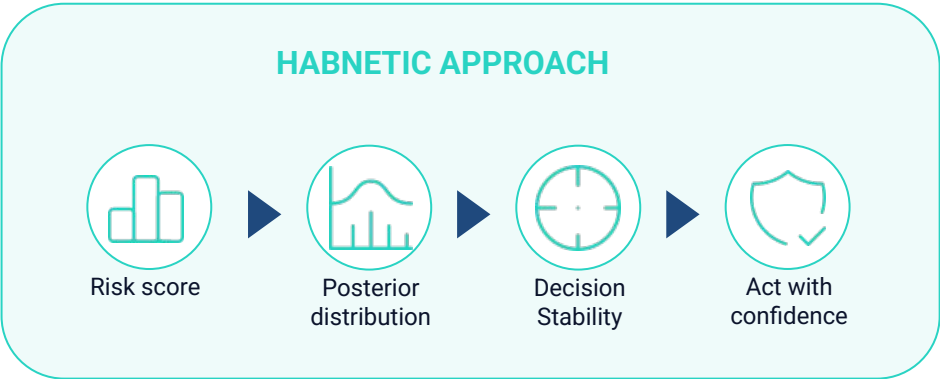
Greater confidence where it matters. Less effort where it doesn't.

From ranking to uncertainty-aware prioritisation

Traditional prioritisation systems produce a ranking and implicitly assume the ranking is reliable.

Habnetic treats prioritisation as a probabilistic object, estimating how stable decisions remain under uncertainty.

The objective is not to eliminate uncertainty, but to identify where decisions remain structurally reliable and where they do not.



TAKEAWAY

The objective is not to eliminate uncertainty.
The objective is to identify where decisions remain reliable and where they do not.

DECISION STABILITY ENABLES



Targeted review



Robust prioritisation



Transferability assessment



Explicit uncertainty communication



Defensible resource allocation



Decision stability



Targeted review



Defensible decisions



Transferable framework